```verilog
`timescale 1ns / 1ps
`default_nettype none
`include "parameters.vh"

module tb_IF;

    // Inputs
    reg clk;
    reg rst_n;
    reg stall;
    reg [31:0] i_Inst;
    reg i_Imem_ack;
    reg [31:0] i_Imm;
    reg [31:0] i_Result;
    reg i_Boj;
    reg i_Jalr;

    // Outputs
    wire [31:0] o_PcD;
    wire [31:0] o_InstrD;
    wire [31:0] o_Iaddr;
    wire o_Imem_stb;

    // Instantiate the Unit Under Test (UUT)
    IF uut (
        .clk(clk),
        .rst_n(rst_n),
        .stall(stall),
        .o_PcD(o_PcD),
        .o_InstrD(o_InstrD),
        .i_Inst(i_Inst),
        .i_Imem_ack(i_Imem_ack),
        .o_Imem_stb(o_Imem_stb),
        .o_Iaddr(o_Iaddr),
        .i_Imm(i_Imm),
        .i_Result(i_Result),
        .i_Boj(i_Boj),
        .i_Jalr(i_Jalr)
    );

    // Clock generation
    always #5 clk = ~clk;

    initial begin
        $display("Starting IF stage AMO test...");
```

```
        // Initialize Inputs
        clk = 0;
        rst_n = 0;
        stall = 0;
        i_Inst = 32'b0;
        i_Imem_ack = 0;
        i_Imm = 32'h80000000;
        i_Result = 32'h00000000;
        i_Boj = 0;
        i_Jalr = 0;

        // Reset the design
        #10 rst_n = 1;
        #10;

        // Test 1: AMOADD.W x5, x6, (x7)
        i_Inst = {5'b00110, 5'b00111, `AMO_ADD, 2'b00, 5'b00101, `AMO}; // rd = x5,
rs2 = x6, rs1 = x7
        i_Imem_ack = 1;
        #10;

        // Test 2: AMOSWAP.W x5, x6, (x7)
        i_Inst = {5'b00110, 5'b00111, `AMO_SWAP, 2'b00, 5'b00101, `AMO};
        #10;

        // Test 3: AMOMIN.W x5, x6, (x7)
        i_Inst = {5'b00110, 5'b00111, `AMO_MIN, 2'b00, 5'b00101, `AMO};
        #10;

        // Test 4: AMOMAXU.W x5, x6, (x7)
        i_Inst = {5'b00110, 5'b00111, `AMO_MAXU, 2'b00, 5'b00101, `AMO};
        #10;

        // Test 5: AMO_LR.W x5, (x7)
        i_Inst = {5'b00000, 5'b00111, `AMO_LR, 2'b00, 5'b00101, `AMO};
        #10;

        // Test 6: AMO_SC.W x5, x6, (x7)
        i_Inst = {5'b00110, 5'b00111, `AMO_SC, 2'b00, 5'b00101, `AMO};
        #10;

        // Test 7: Simulate PC jump (Branch Taken)
        i_Boj = 1;
        i_Imm = 32'h80000100; // New PC target
        #10;
        i_Boj = 0;
```

```
        // Test 8: Simulate Stall (Hazard Scenario)
        stall = 1;
        i_Inst = {5'b00110, 5'b00111, `AMO_OR, 2'b00, 5'b00101, `AMO};
        #10;
        stall = 0;

        // Test 9: Reset Test
        rst_n = 0;
        #10;
        rst_n = 1;
        #10;

        $display("Test completed.");
        $stop;
    end

endmodule
```